

Week 10 Seminar Tasks

Matrices. Part 2

READING

Lecture 8 Notes (available on Blackboard)

Chapter 27. Croft, T and Davison R (2016) *Foundation maths*, 6th ed. Harlow: Pearson.

TASK 1. Singular & Non-Singular Matrices. The inverse of 2×2 Matrix.

See Lectures 8 Notes -Slides 5-10

Establish which of the following matrices have their inverse by any of the applicable techniques, and either explain why an inverse does not exist or calculate it.

$$\mathbf{A} = [0] \quad \mathbf{B} = [-0.5] \quad \mathbf{C} = [2.5] \quad \mathbf{D} = \begin{bmatrix} 1 & 2 \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \quad \mathbf{H} = \begin{bmatrix} 1 & -2 \\ 2 & 4 \end{bmatrix} \quad \mathbf{J} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

TASK 2. The Inverse of 3×3 Matrix

See Lectures 8 Notes -Slides 20-26

For the following 3x3 matrices. We will use an enumeration of the elements starting with a11.

$$\mathbf{K} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\mathbf{L} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

$$\mathbf{N} = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & 5 \\ -1 & 2 & 3 \end{bmatrix}$$

$$\mathbf{O} = \begin{bmatrix} 1 & 3 & -1 \\ -2 & 1 & 2 \\ 0 & 5 & 3 \end{bmatrix}$$

Find for each of these matrices:

- Its *Determinants* using the *diagonal method*
- Matrix of Minors* of its elements, call this matrix **M**
- Matrix of Cofactors* of its elements, call this matrix **C**
- The *Adjoint Matrix* called this matrix **A**
- The *inverse* – $\mathbf{K}^{-1}$, $\mathbf{L}^{-1}$, $\mathbf{N}^{-1}$ and $\mathbf{O}^{-1}$
- Check that the matrices you found are such that $\mathbf{K} \times \mathbf{K}^{-1} = \mathbf{I}$; $\mathbf{L} \times \mathbf{L}^{-1} = \mathbf{I}$; $\mathbf{N} \times \mathbf{N}^{-1} = \mathbf{I}$ and $\mathbf{O} \times \mathbf{O}^{-1} = \mathbf{I}$ where **I** is the *identity matrix* of the same size as *K, L, N, O* correspondently.
- If you find any correlation between any of the given matrices, then try to find the reasoning path to determine how to use it to simplify the inverse calculation. Formulate this correlation as a general property and consider any restrictions.

TASK 3. Message Encryption and Decryption Using Matrix Inversion

See Lectures 8 Notes - Slides 27-35

You are given a 3×3 key matrix K and a message vector M . The message is encrypted by multiplying K with M to produce an encrypted vector E . Your task is to calculate the encrypted message E , find the inverse of K , and decrypt E to recover the original message vector M .

Matrix K :

$$K = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 1 & 5 \\ 3 & 2 & 6 \end{bmatrix}$$

Message Vector M :

$$M = \begin{bmatrix} 8 \\ 15 \\ 7 \end{bmatrix}$$

Task:

1. Encryption: Calculate the encrypted message vector E using K and M .
2. Determinant: Compute the determinant K using the diagonal method to confirm it is invertible.
3. Inverse Matrix:
 - Calculate the matrix of minors for K .
 - Use the minors to form the cofactor matrix.
 - Transpose the cofactor matrix to compute the adjoint matrix.
 - Use the adjoint matrix and determinant to find the inverse matrix K^{-1} .
4. Decryption: Use K^{-1} to decrypt E and recover the original message vector M .